Community Energy (CE) Pilot for El Arenal

Prepared for Red Local de Iniciativas de Conservación, Sustentabilidad y Buen Vivir “El Arenal” and EPM

CoGrid CONSULTING

El Arenal

Project Context & Client Requirements

A shared solution within community, utility and project constraints

Community goals

Lower electricity costs
Greater energy resilience
Community participation
Environmental protection

EPM goals

A viable pilot
Grid stability
Regulatory compliance
A repeatable delivery model

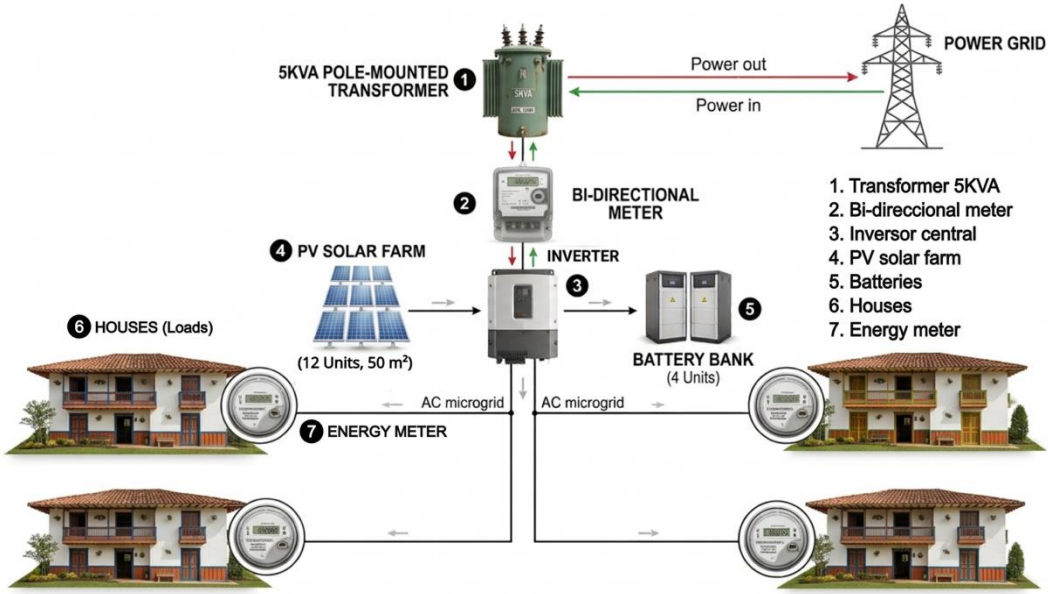
Project constraints

Existing 5 kVA transformer
Grid-connected solution
Short implementation timeline
Current Colombian regulation for CE already connected

Consulting objective: balance both clients' priorities within the project constraints, with an implementation framework that EPM can reuse.

Technical Solution

Shared solar PV and battery storage connected to EPM through the existing transformer and a micro-grid



System specs

System architecture

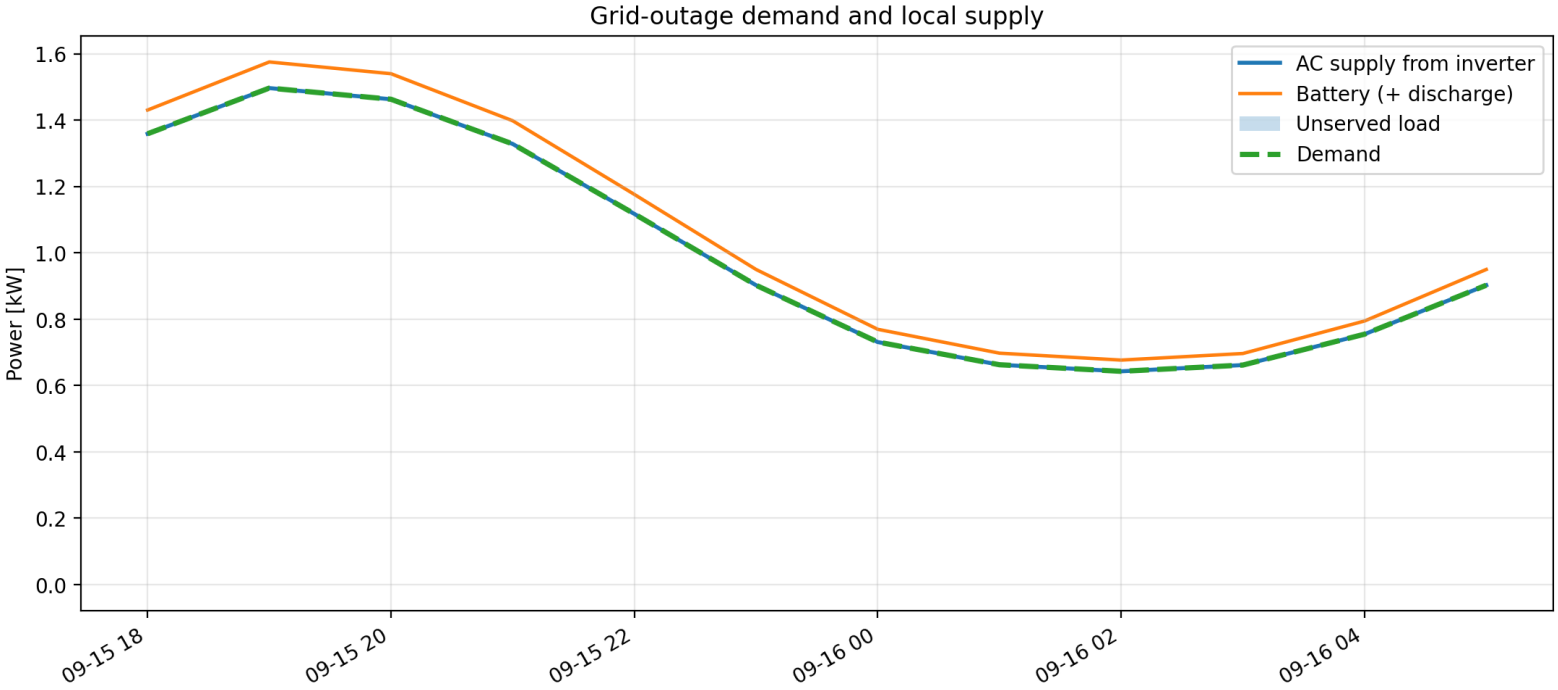
PV Array	5.22 kW
Grid	1,000 kW
Battery	4 Deye 5.12 kWh
Inverter	5 kW
Rectifier	5 kW

Cost summary

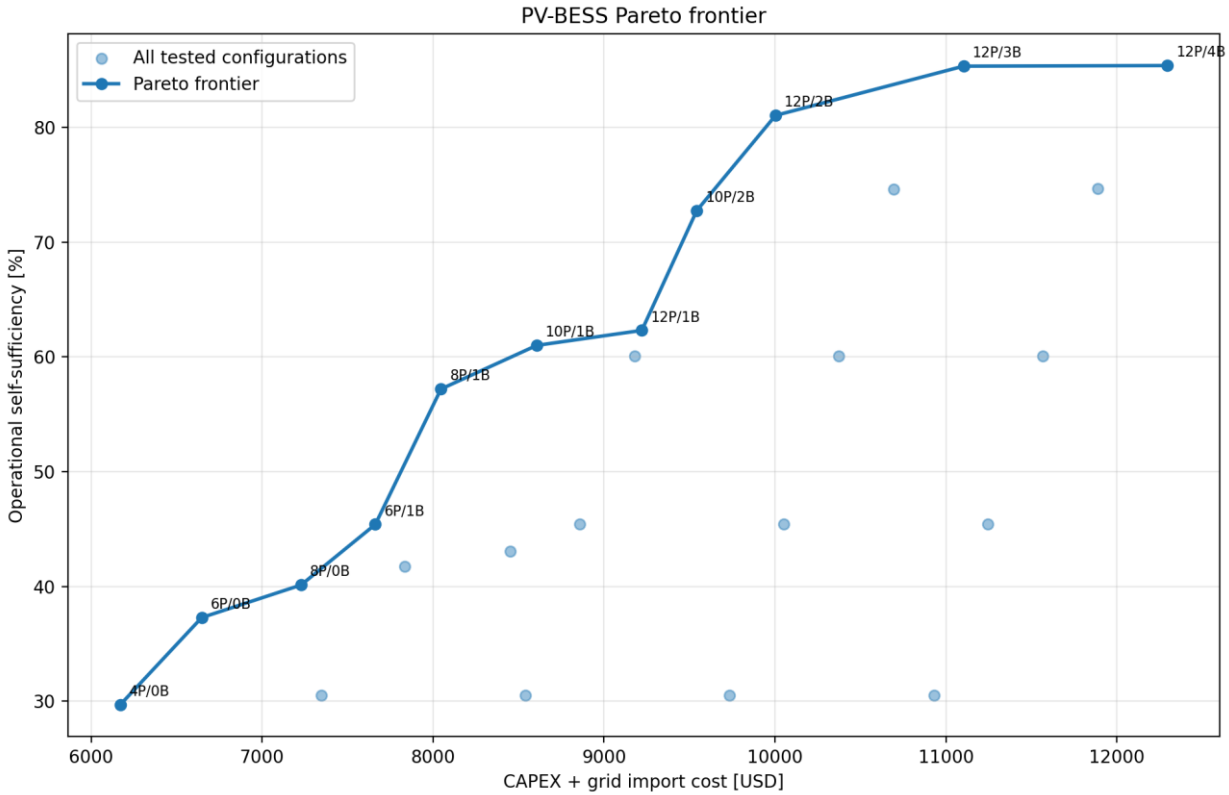
Total net present cost	$ 16,142
Levelized cost of energy	$ 0.166/kWh
Operating cost	$ 327/yr

Team-supplied topology. Final component ratings and model performance remain to be confirmed.

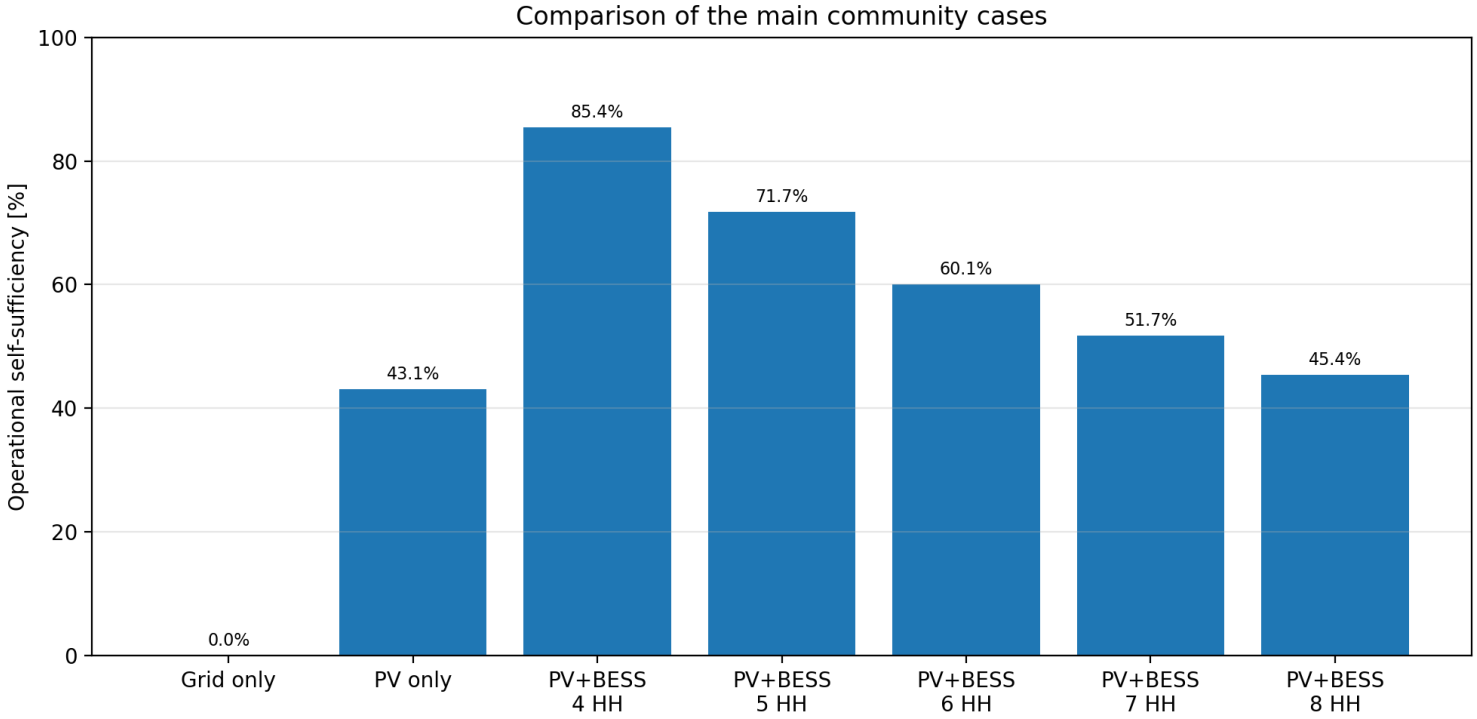
Technical Solution



Technical Solution

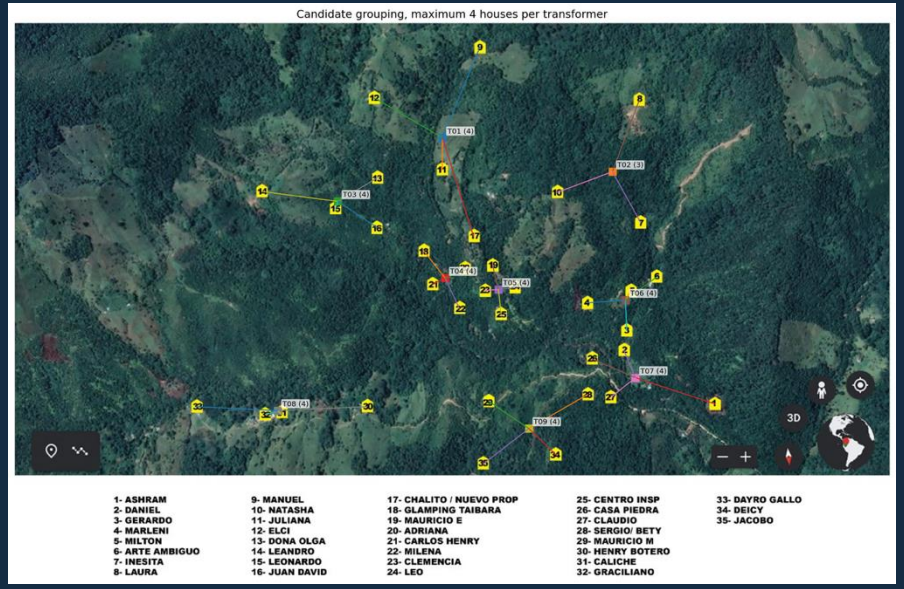


Technical Solution

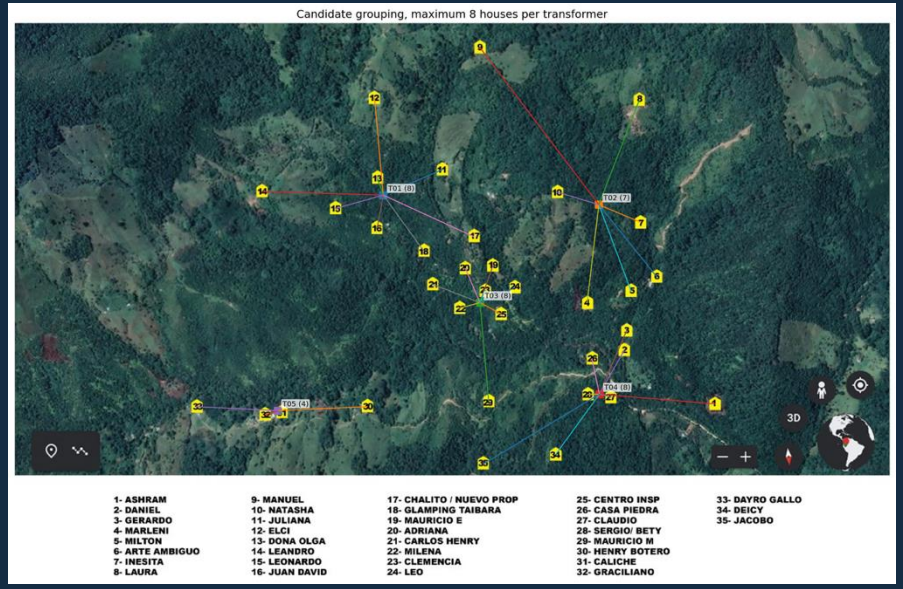


Technical Solution

Spatial distribution analysis



4 houses per transformer

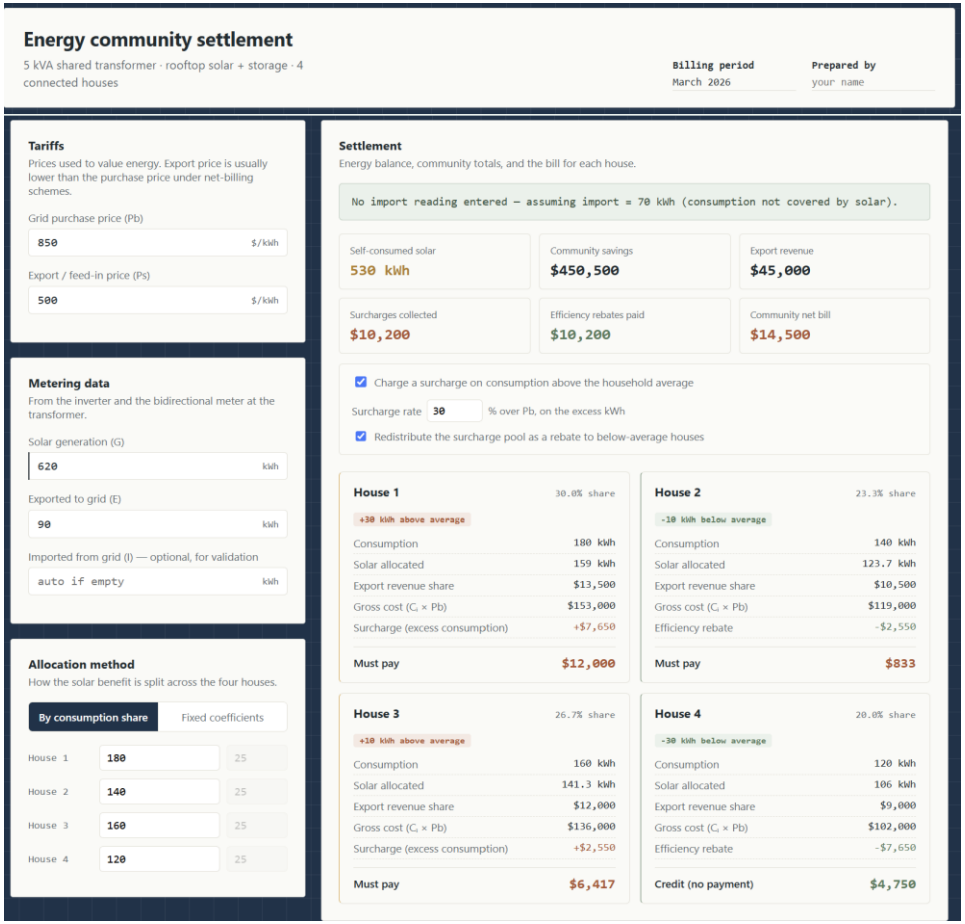


8 houses per transformer

Technical Solution

Energy community billing calculator

End-user platform developed by CoGrid for community energy management.



Pilot implementation cost

5.22 kW Grid-Tied Solar System for 96.15% Energy Savings

RENEW ENERGY AND HELP THE PLANET

Annual Energy Savings



3,241

CO2 Savings



54 Trees Not Cut Down



1,383

Liters of Gasoline

- System designed with 12 photovoltaic modules of 435.00 W to achieve 5.22 kW of power and generate an average of 20.00 kWh per day. Your current average daily consumption is 20.80 kWh.
- One 5.00 kW inverter is proposed to connect the photovoltaic installation to the grid.
- This would represent a 97.09% reduction compared to what you would pay on the current rate.
- System cost per Watt installed: $2.29 USD/Watt + Taxes
- Save the standard 19% VAT on equipment and services. Applies to panels, inverters, wiring, structures, and installation services

Quotation:

Quantity	Name	Description
12	Longi Solar 435W	Longi Solar PV Module 435W Hi MO X6 Explorer
1	Deye Hybrid Inverter 5K 3P	Deye Hybrid Inverter 5K 3P
4	DEYE 5.12 kWh Battery	DEYE 5.12 kWh Battery
12	Installation	Industrial-grade galvanized steel structure for module support. Electrical equipment, CFE (Federal Electricity Commission) paperwork, installation, and commissioning included.
Subtotal		*$11,974.79 USD*
Iva		*$2,275.21 USD*

Delivery time: 25 days
Warranty: 15 years on solar panels, 10 years on inverter, and 3 years on installation.

TOTAL:

$14,250.00 USD

Why This Solution?

Comparison of the client's initial approach and CoGrid's proposed operating design

Design choice	Client's initial proposal: Individual rooftop systems	CoGrid proposed solution: Community microgrid
Installation	A separate solar system for each household	Shared solar PV and batteries serving the community
Ownership and decisions	Each household owns and manages its own system	Community owns shared assets and sets collective rules in the ACE
Coordination with EPM	Household generation connections managed individually	One coordinated project and community energy representative*
5 kVA transformer	Separate inverters need coordination to limit total grid imports / exports	Shared controls manage battery use and net grid imports / exports**
Operation & maintenance	Each household arranges repairs and equipment replacement	One O&M team, maintenance plan and shared replacement fund
Monitoring and data	Separate system records must be combined to assess community impact	Central monitoring records generation, storage and grid exchange
Replication for EPM	Plan equipment, agreements and maintenance for each household	Reuse a community project package: sizing method, ACE, O&M and monitoring

*EPM must approve connection / metering arrangements and transformer operating limits for either option. **end-user app developed by CoGrid

Recommendation: shared ownership and O&M serve El Arenal's priorities; coordinated operation and reusable project documentation serve EPM's.

Recommended Regulatory Path & Operating Model

AGRC fits the pilot's purpose: meeting members' demand and reducing purchased energy

RECOMMENDED NOW

Self-consumption · AGRC

Serve members' demand first.
Surplus can receive regulated remuneration.
Associative legal person under Art. 2.2.9.1.5.

FUTURE OPTION

Energy sales / commercialization

GDC: generation for sale via a generator.
Retail trading adds CREG obligations.
Legal form follows Art. 2.2.9.1.6.

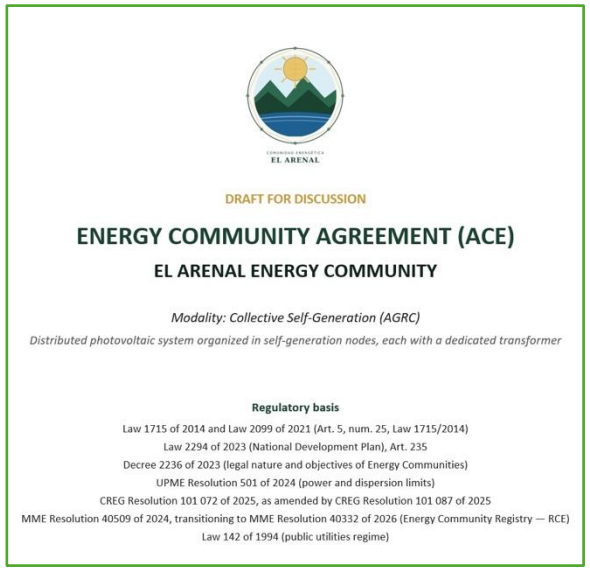
ACE = the community agreement on representation, contributions, decisions, benefit sharing and O&M.



Before commissioning: confirm AGRC connection points, meter ownership and settlement with EPM.
Choose the legal form appropriate to the activity; GDC is not the same as retail commercialization.

Value Proposition

What the pilot is designed to deliver for El Arenal and EPM



DRAFT FOR DISCUSSION

ENERGY COMMUNITY AGREEMENT (ACE)

EL ARENAL ENERGY COMMUNITY

Modality: Collective Self-Generation (AGRC)

Distributed photovoltaic system organized in self-generation nodes, each with a dedicated transformer

Regulatory basis

Law 1715 of 2014 and Law 2099 of 2021 (Art. 5, num. 25, Law 1715/2014)
Law 2294 of 2023 (National Development Plan), Art. 235
Decree 2236 of 2023 (legal nature and objectives of Energy Communities)
UPME Resolution 501 of 2024 (power and dispersion limits)
CREG Resolution 101 072 of 2025, as amended by CREG Resolution 101 087 of 2025
MME Resolution 40509 of 2024, transitioning to MME Resolution 40332 of 2026 (Energy Community Registry — RCE)
Law 142 of 1994 (public utilities regime)

El Arenal

- Lower electricity costs*
- Resilience for priority services*
- Local ownership and shared decisions
- Trained residents to operate
- and maintain community assets
- Skills for productive opportunities

EPM

- Real operating data to validate replication
- A framework to integrate Energy Communities
- Improved relationships with the community
- Local teams able to operate their own assets
- Potential distributed generation and, later,
- virtual power plant (VPP) participation

*Savings and resilience need model and field validation. VPP participation is a future option, subject to technical capabilities, agreements and applicable regulation.

Replication Framework for EPM

A framework for future Comunidades Energéticas projects within EPM's service territory



Monitoring responsibilities

Scope achieved during the hackathon

Community: local asset operation, maintenance and household energy costs.
EPM: grid-interface performance and data for replication. Joint review: pilot outcomes.

Roadmap & Final Recommendation

A staged rollout that uses operating evidence to guide replication



1	2	3	4	5
Pilot	**Monitor**	**Evaluate**	**Optimize**	**Replicate**
Commission safely	Collect operating data	Compare with targets	Improve controls / O&M	Adapt the next site

CoGrid Consulting recommends

Implement El Arenal as a grid-connected AGRC Energy Community with shared solar PV and battery storage, subject to technical and connection validation. Use the pilot to guide replication across EPM's territory.

Questions?